

Java Order of Operations

Priority	Operator(s)	Description
1	()	Parentheses
2	**, Math.pow()	Exponentiation
3	!	Logical NOT
4	*, /, %	Multiplication, Division, Modulus
5	+, -	Addition, Subtraction
6	>, <, >=, <=	Relational / Comparison
7	==, !=	Equality
8	&&	Logical AND
9		Logical OR
10	=, +=, -=	Assignment

Java Control Flow

if

```
if (condition) {  
    // code block  
}
```

if/else

```
if (condition) {  
    // code block  
} else {  
    // code block  
}
```

Ternary if/else

```
variable = (condition) ? valueIfTrue : valueIfFalse;
```

if/else-if

```
if (condition1) {  
    // code block  
} else if (condition2) {  
    // code block  
} else {  
    // code block  
}
```

Switch

```
switch(expression) {  
    case x:  
        // code block  
        break;  
    case y:  
        // code block  
        break;  
    default:  
        // code block  
}
```

for loop

```
for (int i = 0; i < 5; i++) {  
    // code block  
}
```

while loop

```
while (condition) {  
    // code block  
}
```

do-while loop

```
do {  
    // code block  
} while (condition);
```

Java Methods

```
public static <returnType> <methodName>(<parameters>) {  
    // code that runs when the method is called  
    return <someValue> // Note: Don't need 'return' if <returnType> is `void`  
}
```

Java Arrays

Create with values

```
int[] numbers = {10, 20, 30, 40, 50};  
String[] colors = {"red", "green", "blue"};
```

Access / Modify

```
numbers[0]           // first element  
numbers[2] = 99;     // change third element  
numbers.length       // number of elements
```

Create empty (fixed size)

```
int[] numbers = new int[5];
```

for-each loop

```
for (int num : numbers) {  
    // code block  
}
```